Hashing Concepts & Hash Functions Data Structures

> **Definition: Hashing** is a technique that transforms a search key `K` into a table index `h(K)` using a **Hash Function**, enabling average constant time **O(1)** insertion, deletion, and lookup operations in a **Hash Table**.

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1. Detailed Technical Explanation

```
Search
```

1. The Load Factor (**lpha**)

The load factor represents the density of items stored in a hash table of size `m` with `n` keys:

```

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## 2. Common Hash Functions

### 1. Division Method (Modulo Arithmetic)

```
h(k) =

2. Mid-Square Method

1. Square the key: **k^2**.

2. Extr

3. Folding Method

Divide the key digits into equal parts of size **r**, and sum them together (ignoring overflow carry).

- **F

4. Multiplication Method

```
h(k) =
```

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3. Properties of a Good Hash Function

1. **Uniform Distribution:** Distributes keys evenly across all table slots to minimize collisions.

2. **De

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4. Quick Recall Flow

```
Key ->
```